

Edge AI Patent Assessment

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SUMMARY

Conclusion: Label P: TRL 85.0 / Claim 85.0
Key reasons: TRL readiness score 85.0 / Claim breadth score 85.0
Purpose: Edge AI 특허 기술성·시장성 평가 | Keywords on-device, edge ai, federated learning, quantization | CPC G06N, G06F | Period 2022-01-01 ~ 2025-12-31
Top-line: N=2 / M=1 / K=0 | Label P

Recommendations

- FTO 상세 검토 (owner IP 팀, due +7d): 리스크 플래그 대응
- 표준 연계 확인 (owner 표준화 팀, due +14d): SEP 선언 여부 검증
- 데이터 보완 (owner AI 전략팀, due +21d): 결측 지표 5개 채우기

Risks

- 데이터 결측: 출시 전 법률/데이터 보강 필요
- 데이터 결측: 5/10 지표만 확보

Trust meta: missing 50.0% | Executed 2025-10-23T13:27:04.501137+09:00 | Policy v0
MAIN BODY

1. Scope & Method

- scope: {'keywords': 'on-device, edge ai, federated learning, quantization', 'cpc': 'G06N, G06F', 'period': '2022-01-01 ~ 2025-12-31', 'target_market': 'trl'}
- method: {'normalisation': '퍼센타일 → 5점(설계 예정)', 'weights': '기술/시장/경쟁 가중 합산', 'label_rule': 'A/P/M/X 총점 및 FTO 보수 규칙'}

2. Dataset Overview

- documents: 2
- family_size: 1
- quality: {'missing_metrics': ['COUNTRIES', 'GENERALITY', 'HHI', 'LITIGATION_FLAG', 'ORIGINALITY'], 'translation_flags': 'N/A', 'dedup': 'N/A'}

3. Core Metrics

- citations_total: 0
- family: 1
- legal_status: active
- trl_score: 85.0
- claim_score: 85.0

4. Edge Fitness

5. Competitive & Standards

- sep: declared

6. Risk Profile

- data_gaps: COUNTRIES, GENERALITY, HHI, LITIGATION_FLAG, ORIGINALITY

7. Ranking & Labelling

- total_score: 74.09210526315789
- label: P
- distribution: {'A': 0, 'P': 0, 'M': 0, 'X': 0}

8. Decision Cards

- title=US20230012345; family=1; owner=Edge AI Accelerator for Wearables;
jurisdictions=[]; decision=Adopt; signals=['TRL readiness score 85.0', 'Claim breadth
score 85.0']; risks=[{'item': '데이터 결측', 'impact': '출시 전 법률/데이터 보강 필요'}, {'item': '데이터
결측', 'impact': '5/10 지표만 확보'}]

9. Conclusion

- TRL readiness score 85.0
- Claim breadth score 85.0

REFERENCE

[R1] NBER Working Paper 7741 - <https://www.nber.org/papers/w7741>
[R2] Google Patents - <https://patents.google.com/patent/US20230012345>
[R3] WIPO IP Indicators - <https://www.wipo.int/web-publications/world-intellectual-property-indicators-2024-highlights/en/patents-highlights.html>
[R4] USPTO Official Gazette -
<https://www.uspto.gov/web/offices/com/sol/og/2012/week52/TOCCN/item-80.htm>
[R5] WIPO Magazine - Freedom to Operate - <https://www.wipo.int/en/web/wipo-magazine/articles/ip-and-business-launching-a-new-product-freedom-to-operate-34956>
[R6] ETSI SEP List - https://www.etsi.org/deliver/etsi_sr/000300_000399/000314/02.32.01_60/sr_000314v023201p.pdf
[R7] DOJ / FTC Guidelines - <https://www.justice.gov/atr/herfindahl-hirschman-index>
[R8] NBER Working Paper 8498 - <https://www.nber.org/papers/w8498>
[R9] Local PDF - data\pdfs\WO2018097365A1.pdf
[R10] Local PDF - data\pdfs\SWTRLs.pdf

APPENDIX

purpose: Edge AI 특허 기술성·시장성 평가
pipeline: PDF → Chunk → Evidence RAG → 고정 메트릭
formulae: 가중합 = $\sum(\text{metric} \times \text{weight})$, 결측 시 가중 재분배
label_policy: Adopt/Prototype/Monitor/Archive (설계 중)
device_thresholds: latency_ms=50, power_mw=600, memory_mb=256
sensitivity: contributions={'trl': 0.2631578947368421, 'claim': 0.2631578947368421, 'legal': 0.10526315789473684, 'family': 0.13157894736842105, 'foreign': 0.10526315789473684, 'renewal': 0.07894736842105263, 'std': 0.05263157894736842}, redistributed=True, missing_keys=['generality', 'originality'], fto_penalty=0.0
risk_policy: 데이터 결측/표준 미확인 시 보수적 하향
execution_meta: timestamp=2025-10-23T13:27:04.501137+09:00, version=v0

EVIDENCE

TRL [Local RAG]

TRL level: 8.5

Example progression: prototype -> pilot -> field test -> deployed production services.

TRL 5 corresponds to validating a software prototype in a relevant environment. TRL 6 and TRL 7 involve field test pilots with partially deployed systems. TRL 8 and TRL 9 indicate production-ready and fully deployed software in an operational environment.

CLAIMS [Local RAG]

Independent claims: 1, Avg tokens: 177.0

API_METRICS [Local RAG]

SCORE [Local RAG]